

Functional Requirements

Functional requirements describe what the system should do.

1. User Registration & Login

- The system should allow users to register and log in securely.
- Users can access features after successful authentication.

2. Building Information Input

- Users should be able to enter building details such as:
 - Building type (home, school, hospital, office)
 - Construction material
 - Age of the building
 - Location and surroundings

3. Risk Level Prediction

- The system should analysed the provided data.
- It should classify the building as:
 - Safe
 - Moderate Risk
 - High Risk

4. Disaster Risk Assessment

- The system should assess risks for disasters like earthquakes, floods, and fires.
- It should consider structural and environmental factors.

5. Safety Precautions & Guidelines

- The system should provide personalized safety instructions.
- Precautions should vary based on the risk level.

6. Safe Place Recommendation

- The system should recommend nearby:
 - Emergency shelters
 - Hospitals

- Evacuation areas
- Recommendations should be location-based.

7. Emergency Alerts & Guidance

- The system should guide users on what actions to take during emergencies.
- It should support quick decision-making.

8. Map & Location Support

- The system should display safe places on a map.
- GPS/location services should be used for accuracy.

9. Admin Management

- Admin should manage:
 - Disaster data
 - Safe place information
 - System updates

10. User Feedback & Support

- Users should be able to give feedback or report issues.

Non-Functional Requirements (Detailed)

Non-functional requirements describe how well the system should perform.

1. Performance

- The system should respond quickly to user requests.
- Risk prediction results should be generated in minimal time.

2. Scalability

- The system should handle a large number of users during disasters.
- It should scale without performance issues.

3. Reliability

- The system should function continuously during emergency situations.
- It should provide accurate and consistent results.

4. Security

- User data should be protected using encryption.
- Secure authentication should be implemented.

5. Usability

- The system should be easy to use for all age groups.
- The interface should be simple and user-friendly.

6. Availability

- The system should be available 24/7.
- Downtime should be minimal, especially during disasters.

7. Privacy

- User personal and location data should not be shared without consent.
- Data handling should follow privacy standards.

8. Portability

- The system should work on multiple devices like mobile phones, tablets, and desktops.

9. Maintainability

- The system should be easy to update and maintain.
- New disaster types or safety rules can be added easily.

10. Accuracy

- Predictions and recommendations should be as accurate as possible based on data.